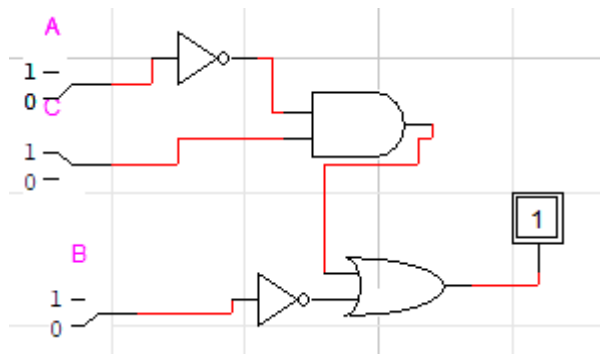


Task 1:

TASK :: 1

AB \ C	0	1
00	1	1
01		1
11		
10	1	1

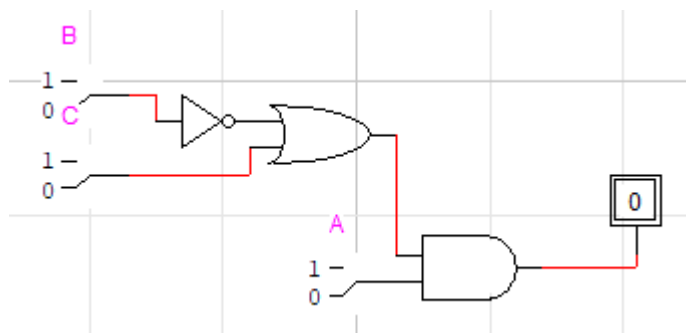
$\bar{B} + A'C$

**Task 2:**

TASK :: 2

AB \ C	0	1
00	0	0
01	0	0
11	0	
10		

$AC + A\bar{B}$
 $= A(\bar{B} + C)$



Task 3:

A	B	C	$\bar{A}$	$\bar{B}$	$\bar{C}$	$\bar{A}\bar{B}C$	$A\bar{B}\bar{C}$	ABC	$\bar{A}\bar{B}C + A\bar{B}\bar{C} + ABC$
0	0	0	1	1	1	0	0	0	0
1	0	0	0	1	1	0	1	0	1
0	1	0	1	0	1	0	0	0	0
0	0	1	1	1	0	1	0	0	1
1	1	0	0	0	1	0	0	0	0
1	0	1	0	1	0	0	0	0	0
0	1	1	1	0	0	0	0	0	0
1	1	1	0	0	0	0	0	1	1

Task 4:

A	B	C	$\bar{A}$	$\bar{B}$	$\bar{C}$	$A+B+C$	$A+\bar{B}+C$	$A+\bar{B}+\bar{C}$	$\bar{A}+B+C$	$\bar{A}+\bar{B}+C$	$(A+B+C)(A+\bar{B}+C)(A+\bar{B}+\bar{C})(\bar{A}+B+C)(\bar{A}+\bar{B}+C)$
0	0	0	1	1	1	0	1	1	1	1	0
0	0	1	1	1	0	1	1	1	1	1	1
0	1	0	1	0	1	1	0	1	1	1	0
1	0	0	0	1	1	1	1	1	1	1	1
1	1	0	0	0	1	1	1	1	1	0	0
1	0	1	0	1	0	1	1	1	0	1	0
0	1	1	1	0	0	1	1	0	1	1	0
1	1	1	0	0	0	1	1	1	1	1	1

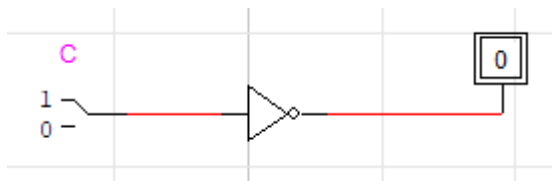
Task 5:

SOP: $A'B'C + AB'C' + ABC' + ABC$
POS: $(A+B+C)(A+B+\bar{C})(A+\bar{B}+C)(\bar{A}+B+C)$

Task 6:

AB \ CD	00	01	11	10
00	1	1		
01	1	1		
11	1	1		
10	1	1		

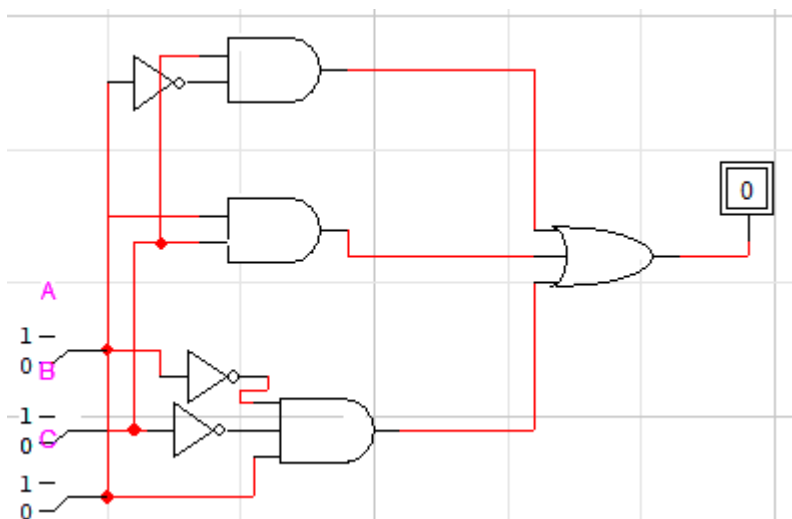
$= \bar{C}$

**Task 7:**

Task 7:-

AB \ C	00	01	11	10
00		1		
01	1			
11	1	1		
10				

$B\bar{C} + AB + \bar{A}\bar{B}C$



Task 8:

Task 8:-

$$\bar{B}\bar{C} + A\bar{B} + AB\bar{C} + A\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}D + A\bar{B}cD$$

$$= A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + A\bar{B}cD \\ + A\bar{B}C\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + AB\bar{C}D + \\ AB\bar{C}\bar{D} + A\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}D + A\bar{B}c\bar{D}$$

$$= \overset{9}{A\bar{B}\bar{C}D} + \overset{9}{A\bar{B}C\bar{D}} + \overset{1}{\bar{A}\bar{B}\bar{C}D} + \overset{0}{\bar{A}\bar{B}C\bar{D}} + \\ \overset{11}{A\bar{B}C\bar{D}} + \overset{10}{A\bar{B}C\bar{D}} + \overset{13}{A\bar{B}\bar{C}D} + \overset{12}{A\bar{B}C\bar{D}}$$

$$f(A, B, C, D) = \sum m(0, 1, 9, 8, 10, 11, 12, 13)$$

AB \ CD	00	01	11	10
00	1	1	0	0
01	0	0	0	0
11	1	1	0	0
10	1	1	1	1

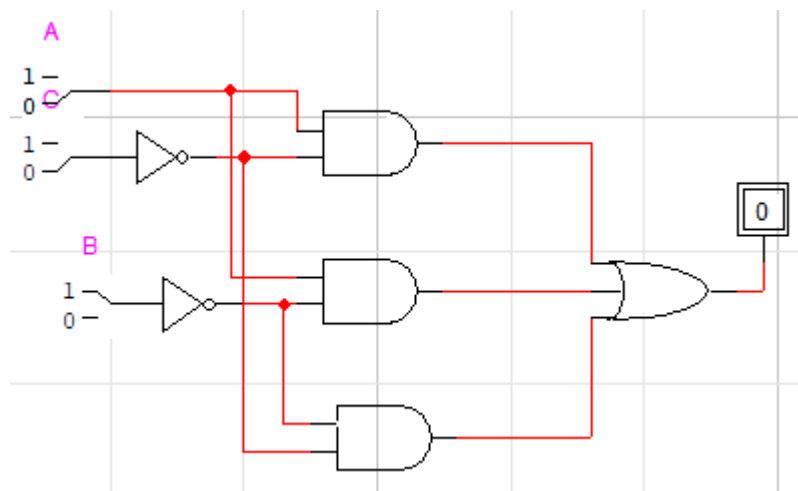
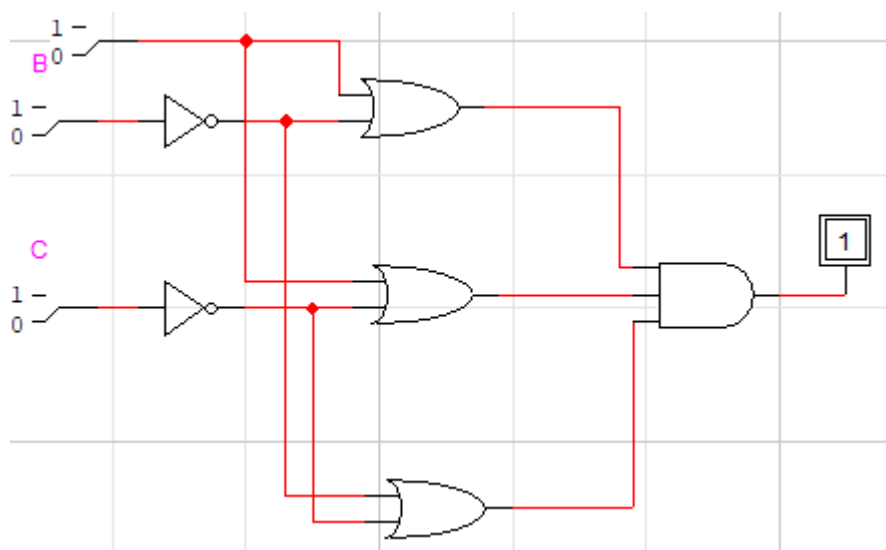
$$A\bar{C} + A\bar{B} + \bar{B}\bar{C}$$

A	B	C	$\bar{A}$	$\bar{B}$	$\bar{C}$	$A\bar{C} + A\bar{B} + \bar{B}\bar{C}$
0	0	0	1	1	1	1
0	0	1	1	1	0	0
0	1	0	1	0	1	0
0	1	1	1	0	0	0

1	0	0	0	1	1	1
1	0	1	0	1	0	1
1	1	0	0	0	1	0
1	1	1	0	0	0	0

$$\text{POS: } (A + \bar{B})(A + \bar{C})(\bar{B} + \bar{C})$$

SOP:

**POS:****Task 9:**

Task 9:-

(a) $AB \backslash C$

	0	1
00	1	
01		1
11	1	1
10		

(b) $AB \backslash C$

	0	1
00	1	1
01	1	
11		1
10	1	1

(c) $AB \backslash CD$

	00	01	11	10
00	1	1		
01	1	1	1	1
11				
10		1	1	

(d) $AB \backslash CD$

	00	01	11	10
00	1			1
01	1	1		1
11	1	1		1
10	1		1	1

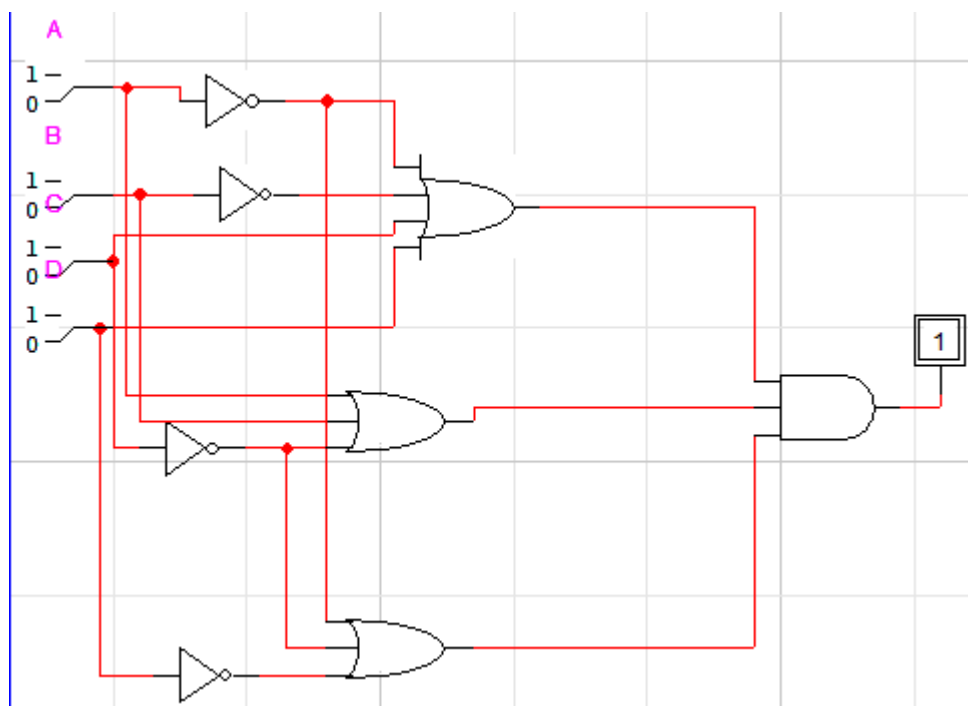
Task 10:

Task 10:-

$$B + \bar{A}C + A\bar{C}D$$
Task 11:

Task 11:-

$AB \backslash CD$	00	01	11	10
00			0	0
01				
11	0		0	
10			0	

$$(\bar{A} + \bar{B} + C + D)(A + B + \bar{C})(\bar{A} + \bar{C} + \bar{D})$$


Task 12:

Task 12:-

AB \ CD	00	01	11	10
00	1	0	0	0
01	0	1	1	1
11	0	1	1	1
10	1	0	1	1

$$(B + C + \bar{D})(A + B + \bar{C})(\bar{B} + C + D)$$

$$F(A, B, C, D) = \sum m(1, 2, 3, 4, 9, 12)$$

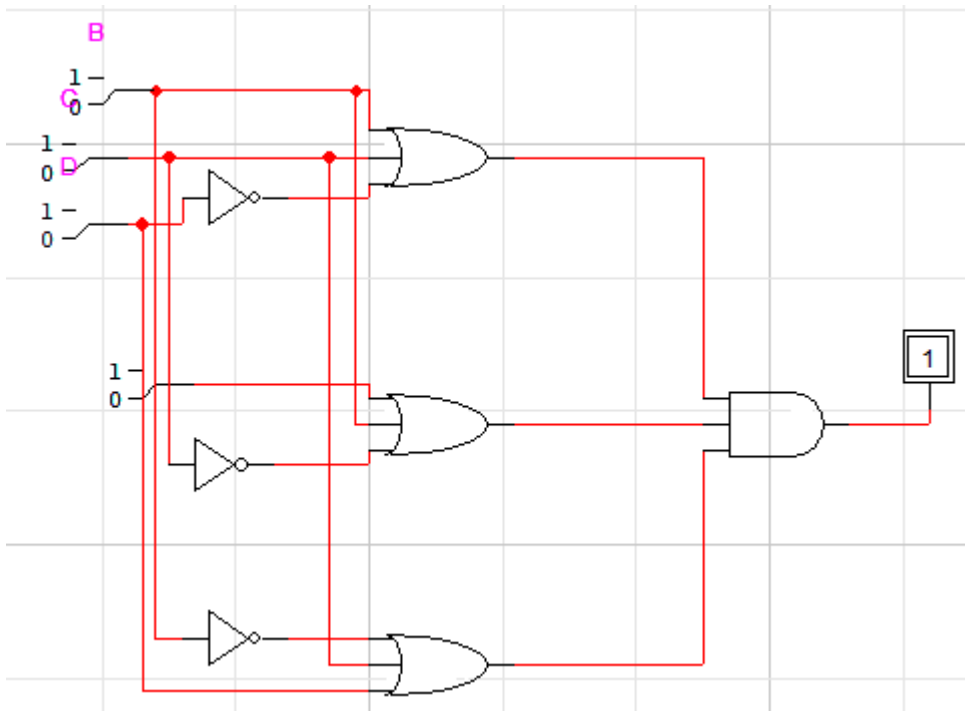
Standard SOP:-

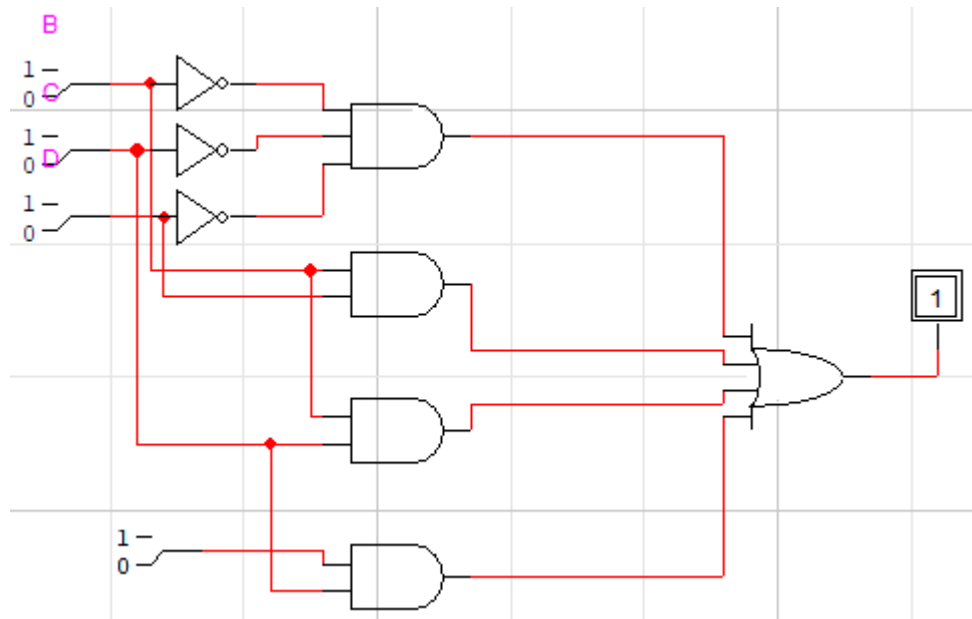
$$F(A, B, C, D) = \sum m(0, 5, 6, 7, 8, 10, 11, 13, 14, 15)$$

$$F = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} + \bar{A}BC\bar{D} + \bar{A}BCD + A\bar{B}\bar{C}\bar{D} + A\bar{B}C\bar{D} + A\bar{B}CD + AB\bar{C}\bar{D} + AB\bar{C}D + ABC\bar{D} + ABCD$$

Minimum SOP:-

$$\bar{B}\bar{C}\bar{D} + BD + BC + AC$$

POS:**SOP:**

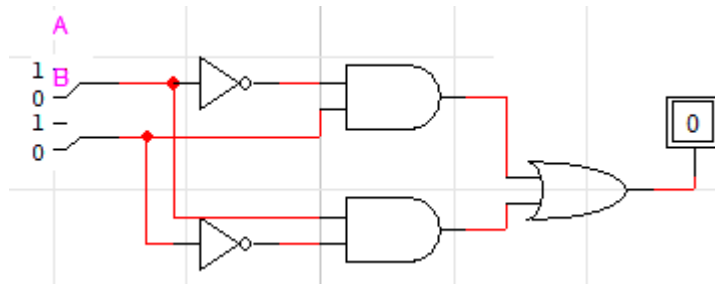
**Task 13:**

1.

1)

A \ B	0	1
0		1
1	1	

 $\bar{A}B + A\bar{B}$

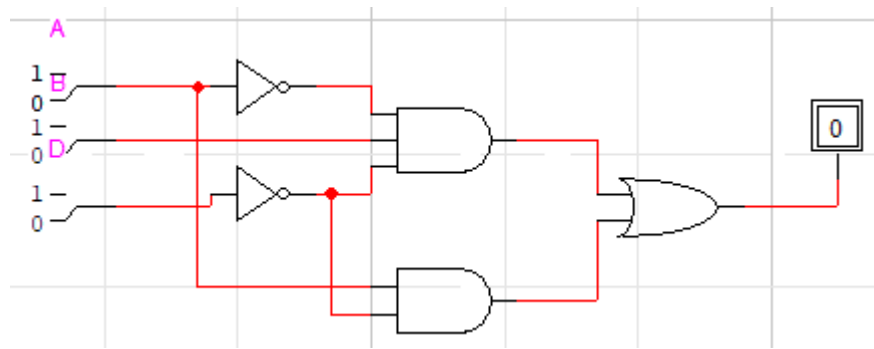


2.

2)

AB \ CD	00	01	11	10
00				
01	1			1
11	1			1
10	1			1

 $\bar{A}B\bar{D} + A\bar{D}$



3.

3. AB/CD

	00	01	11	10
00	1	1	1	1
01		1	1	
11		1	1	
10	1	1	1	1

$\bar{B} + D$

